

A Fixed-Time Language-Model Search That Adopted Nothing: Twenty-Eight Closed Directions, One Adoption, and Its Retraction by Disjoint-Seed Verification

vibeautoresearch_reimplementation
reporting scope walltime_5min_h200

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Abstract

This campaign searched for architectural and optimisation improvements to a byte-level language model under a fixed 300.000000s wall-clock budget on a single H200, scoring held-out `val_bpb` on a pinned validation shard. Over roughly three weeks and 388 recorded runs it closed **28 directions**, produced **one adoption**, and **retracted that adoption one block later** when it was verified on seeds disjoint from those used to select it. The retraction is the campaign’s principal result: the effect proved real (-0.002139 pooled over 20 pairs, $t = -7.18$, 20/20 seeds favouring treatment) but *sub-gate*, and the margin that had carried it over the adoption floor is best explained as selection bias. The campaign’s standing state is **no adopted result**, and its registered scope baseline is flagged **contaminated** after a session in which 45 of 48 runs completed ~ 1147 optimizer steps against an expected ~ 2000 under GPU contention. We report the ledger, the retraction in full, a provenance grading of every headline figure, and the conditions under which comparison with sibling campaigns is and is not meaningful.

1 The frame

All results are scored on one metric under one frame. Deviations from it are not comparable and are marked as such throughout.

Element	Value
Scope identifier	<code>walltime_5min_h200</code>
Stopping rule	<code>STOP_MODE=time, TIME_BUDGET=300</code>
Hardware	one NVIDIA H200, one run per GPU
Objective	held-out <code>val_bpb</code> , lower is better
Validation data	pinned shard 6542
Training data	shards 1–10, frozen
Context length	2048
Typical step count	~ 2000 optimizer steps per run
Adoption gate	paired mean must clear 0.002614 (2σ)
Funnel	$1 \rightarrow 3 \rightarrow 6 \rightarrow 10$ seeds

Table 1: The reporting scope. The gate and the funnel are preregistered.

Because the budget is wall-clock rather than tokens or FLOPs, throughput is part of the objective: a change that costs optimizer steps must repay them in quality. This coupling is central to Section 3.

2 Ledger

Quantity	val_bpb	Provenance	Status
Naive baseline (day-one reference)	1.030100	reconciled, $n = 10$	current
Best historically observed	0.927183	direct-SSH, weak	best-known, <i>not</i> adoption-grade
Best $n \geq 10$ paired mean	0.929541	direct-SSH, weak	best-known, <i>not</i> adoption-grade
Registered scope baseline	0.934930	contaminated	rejected
Old-config reproduction (new host)	0.932028	snapshot, $n = 10$	reproduction, not a SOTA
Best measured configuration	0.929508	E28, $n = 10$	retracted adoption

Table 2: Every headline figure with its provenance grade. Two of the three best numbers are disowned by the campaign’s own records.

The improvement from the day-one reference to the best observed configuration is $1.030100 - 0.927183 = 0.102917$, or 10.000000%. That figure is *historical*: it was accumulated early from a small number of large mechanisms (hashed n -gram memory at 2^{20} , **swin4**, fp8, FA3 varlen document masking), and the campaign has not adopted anything since.

2.1 Adopted snapshots

Six configurations were formally adopted over the campaign’s history:

1. **sota_reoptimized_ffn4_mlr03**
2. **sota_reoptimized_final**
3. **sota_ngram_2p20**
4. **sota_swin4**
5. **sota_fp8_ngram**
6. **sota_fa3_varlen_docmask** — paired $n = 10$, mean -0.005222 , 10/10 same sign, $t = -10.20$

3 The contaminated baseline

The registered scope baseline of 0.934930 ($n = 10$) was measured during a session in which **45 of 48 runs completed ~1147 optimizer steps against an expected ~2000** for the same 300.000000s budget. The cause was GPU contention from another tenant.

The measured penalty was $+0.005390$ relative to the same configuration measured clean. Note the magnitude: contention cost roughly **twice** the campaign’s entire adoption gate. In a wall-clock frame, a co-tenant does not add noise — it removes optimizer steps, and the metric follows.

The baseline was downgraded to **pending** in **reconciliation.json** v32 and **authorize-run** is correctly blocked in this scope until a genuine ≥ 10 -seed remeasurement replaces it.

A second instance appears in the 2026-07-30 reproduction: seeds 42 and 43 were **quarantined** after another tenant saturated their GPUs mid-run (step time $150 \rightarrow 336.000000$ ms, MFU $19\% \rightarrow 8.6\%$, ~ 956 of ~ 2000 steps). They scored 0.989000 and 0.995000 and were discarded rather than averaged in. Had they been included, the reproduction’s mean would have been corrupted by approximately 0.012000 — five gates — by two runs out of ten.

4 The adoption, and its retraction

4.1 E28: the campaign’s first adoption

After **28 closed directions**, block 48 adopted DEPTH=6 (dim 640 via ASPECT_RATIO) with MATRIX_LR=0.03, against frozen defaults DEPTH=8 (dim 768) and MATRIX_LR=0.04.

Statistic	Value
Paired mean	−0.002715
Adoption gate (2σ)	0.002614
Gates cleared	1.040000
Clean pairs	10 / 10
Seeds favouring treatment	10 / 10
t	−5.870000
95% CI lower	−0.003761
95% CI upper	−0.001669
In-tranche control mean	0.932223
Treatment mean	0.929508

Table 3: E28 as recorded. The preregistered rule is a point-estimate rule and it is satisfied.

The record itself flags three reservations at the time of adoption, before any verification was run:

- **The margin is thin.** The point estimate exceeds the floor by 4.000000%, and the upper end of the 95% CI (−0.001669) lies *below* the floor in magnitude, so the true effect may not clear.
- **Against the registered baseline** (0.931857) rather than the in-tranche control, the delta is $-0.002349 = 0.90$ gates, which does *not* clear. Both numbers were recorded so the discrepancy was visible rather than buried.
- **The prediction was that it would fail.** The launcher recorded an expected ~ 0.88 gates. The result came in 0.000415 above the sum of its parts (−0.002002 for depth 6 at $n = 12$; −0.000298 for the LR at $n = 4$), and that super-additivity was explicitly logged as *measured, not explained*.

4.2 E29: verification on disjoint seeds

The adoption was verified on seeds 52–61, disjoint from every arm the campaign had ever run.

Tranche	Seeds	Mean	Gates	n
E28 (original)	42–51 (<i>include the selection seeds</i>)	−0.002715	1.040000	10
E29 (verification)	52–61 (<i>disjoint</i>)	−0.001562	0.600000	10
Pooled	42–61	−0.002139	0.820000	20

Table 4: Neither the verification nor the pooled estimate clears the 0.002614 floor. The adoption was withdrawn.

4.3 Diagnosis: winner’s curse, predicted in advance

The configuration was *selected* from sweeps run on seeds 42–45 (depth 6 from E10; `MATRIX_LR=0.03` from E26/E27) and then *evaluated* on seeds 42–51, which contain all four selection seeds. Selection and evaluation on overlapping draws inflates the estimate.

This was named in the reading log **one block before the verification ran**, with a predicted regression toward the additive estimate of ~ 0.88 gates. The measured regression was +0.001153, from 1.04 gates to 0.60 — close to the prediction.

The 0.000415 super-additive residual that carried E28 over the floor, recorded at the time as *measured, not explained*, is now best explained as selection bias.

4.4 What survives

The effect is real and it is sub-gate. Pooled over 20 pairs: -0.002139 , $t = -7.18$, 20/20 seeds better, 95% CI $[-0.002762, -0.001515]$ — excludes zero, does not reach the floor.

Depth 6 with `MATRIX_LR=0.03` genuinely beats the frozen defaults by about 0.002100 `val_bpb`; the gate asks for 0.002614. The snapshot is retained as the best-known configuration with `repro.json` marked retracted. **It is not a SOTA.**

5 Comparison with sibling campaigns

Two sibling campaigns optimise the same objective on the same host against the same validation shard. Comparison is meaningful only where the model and the batch size match.

Campaign	Best	val_bpb	Model	Tokens/step	Grade
reimplementation	best-known	0.927183	dim 768	2^{18}	weak provenance
v4	promoted	0.969952	dim 512	2^{19}	single run
v1	SOTA	0.962288	dim 512	2^{19}	confirmation, $n = 7$ paired

Table 5: The two campaigns at dim 512 / 2^{19} are directly comparable to each other. Neither is directly comparable to this one.

Two observations follow, and the second is the more useful.

Step counts are not a unit of work. This campaign runs ~ 2000 optimizer steps per 300.000000s frame against ~ 975 for the dim-512 campaigns, which invites the reading that it is twice as fast. At 2^{18} versus 2^{19} tokens per step, the throughputs are 524.000000 million and 511.000000 million tokens respectively — within 3.000000%. The step-count difference is entirely batch size.

The width–batch interaction is unexplored on both sides. This campaign operates at dim 768 with batch 2^{18} . Campaign v1 tested dim 768 *at batch* 2^{19} and measured $+0.078976$ — one of its largest negatives — while never varying `TOTAL_BATCH_SIZE` in 189 experiments. Those two facts are jointly consistent only if width and batch size interact strongly. Neither campaign has probed that diagonal.

6 Discussion

Three properties of this campaign’s record are worth stating plainly, because they are unusual and they are what make the negative result credible.

The verification was run against interest. E29 was executed on disjoint seeds immediately after an adoption the campaign had every incentive to keep — its first in 28 directions. The retraction was published one block later.

The failure mode was predicted before it was measured. The reading log named the overlap between selection and evaluation seeds, and estimated the regression, before E29 ran. The measured regression matched. A campaign that can predict its own errors in advance is calibrated in a way that a campaign reporting only successes cannot demonstrate.

Contamination is treated as invalidity, not as noise. Both the rejected baseline and the two quarantined seeds were removed rather than averaged. In a wall-clock frame this is not fastidiousness: contention removed roughly half the optimizer steps in the affected runs and cost 0.005390 `val_bpb`, about twice the adoption gate. Averaging contaminated runs into a mean does not add variance — it moves the mean.

The cost. The same discipline produces one adoption in three weeks, and then withdraws it. The campaign’s headline 10.000000 % improvement is historical, earned early from a handful of large mechanisms, and its recent record is a plateau punctuated by a retraction. Whether that is a well-calibrated search or an over-tight gate is not settled by the record: a floor of 0.002614 rejects a real, 20/20-consistent effect of 0.002139, and no analysis in the campaign asks whether the floor is correctly placed.

7 Reproducibility

Artefact	Location
Authoritative operating baseline	<code>research/setup/reconciliation.json</code>
Per-adoption code and config	<code>research/sota_snapshots/*/repro.json</code>
Human-readable ledger	<code>SOTA_LOG.md</code>
Retracted adoption	<code>research/sota_snapshots/sota_depth6_lr03_20260801/</code>
Old-config reproduction	<code>research/sota_snapshots/repro_oldconfig_newhost_20260730/</code>
Run-level records	<code>results.tsv</code> , <code>campaign_log.jsonl</code>

Reproduction configuration (2026-07-30, old config on new host): `WINDOW_PATTERN=TTTL`, `DEVICE_BATCH_SIZE=TOTAL_BATCH_SIZE=147456`, `MATRIX_LR=0.04`, FA3 + varlen. Clean $n = 10$: mean 0.932028, sd 0.001032, best 0.930853.

Provenance caveat. Every tranche in this campaign is recorded as direct-SSH behind the probe gate with no `authorize-run` binding, and the campaign batch is marked **quarantined** on provenance even where the science funnel is satisfied. Figures in Table 2 marked *weak* should be read as observed, not as established.